

**Proposed Data Centre
Development at No. 2-10,
Tai Yuen Street, Kwai
Chung, N.T.**

Technical Specification for
Structural Concrete

Revision: 0

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1.0 Structural Concrete Works

1.1 Concretor - General

1.1.1 Scope

This section gives the requirements for all structural concrete in building and civil engineering works, except where otherwise modified or amended within the Contract.

1.1.2 Building Regulations and Local Code of Practice

Where the structural concrete is to be used in any building works in Hong Kong the concrete shall comply with all relevant legislation, Codes of Practice, Practice Notes, except where more stringent requirements are specified herein. Particular attention is to be drawn to the following:

- a) Hong Kong Building (Construction) Regulations 1990,
- b) Code of Practice for Structural Use of Concrete 2013,
- c) Code of Practice for Fire Safety in Building 2011 with its corrigenda,
- d) Code of Practice for Precast Concrete Construction 2003,
- e) Hong Kong Construction Standard CS1:2010 for Testing Concrete,
- f) Hong Kong Construction Standard CS2:2012 for Steel Reinforcing Bars for the Reinforcement of Concrete,
- g) Hong Kong Construction Standard CS3:2013 for Aggregates for Concrete
- h) Practice Notes for Authorized Persons and Registered Structural Engineers (PNAP),
- i) Practice Notes for Registered Contractors (PNRC),

1.1.3 Authoritative Standards and Codes of Practice

- a) The following authoritative documents are referred to the latest edition hereinafter:
- b) British Standard Specifications as published by the British Standards Institution, 2 Park St., London W.1., England (abbreviated in the text to BS).

<u>BS</u>	<u>Title</u>
12:Pt.2	Portland cements (ordinary and rapid hardening).
812	Testing aggregates.
882	Aggregates from natural sources for concrete.
1014	Pigments for Portland cement and Portland cement products.
1370	Low heat Portland cement.
1881	Methods of testing concrete
2499	Hot applied joint sealants for concrete pavements.
3148	Methods of tests for water for making concrete.
3892	Pulverised fuel ash for use as a cementitious component as structural Concrete
4466	Scheduling, dimensioning, bending and cutting of steel reinforcement for concrete.
4483	Steel fabric for the reinforcement of concrete.
5075	Concrete admixtures.
5328	Methods for specifying concrete, including ready mixed concrete.

- c) The Contractor must understand thoroughly any such requirements stated by the foregoing Codes and Regulations before tendering. He shall also construct the work in strict accordance with this Specification, working drawings and written instructions issued by the Engineer.
- d) If the Contractor wishes to substitute any other authoritative standard of code of practice for any listed above he shall submit details of any such together with two complete copies to the Engineer for approval. Approval for such substitution shall be at the discretion of the Engineer and will not be given unless the Engineer considers the proposed standard or code of practice will give a quality of finished work equal to or better than the specified standard.
- e) All insitu concrete shall be in accordance with The Building (Construction) Regulations 1990 and Code of Practice for the Structural Use of Concrete 2013, except where more stringent requirements are specified in the drawings.
- f) The Engineer reserves the right to check the work constructed by the Contractor, and this setting out in such cases, and at such times he may deem fit. There is however no duty on his part to make such checks and any omission by him to observe errors shall not relieve the Contractor of his responsibilities in these respects.

1.1.4 Bending Schedules

- a) Where the Engineer has prepared bar bending schedules these will be prepared generally in accordance with BS 4466. The contractor should check these against the drawings before any cutting, bending or construction involving the schedules is started. Any discrepancy should be reported immediately to the Engineer for his clarification. The Contractor shall be responsible for any delays for additional work caused solely by his failure to check the schedules.

1.1.5 Preliminary Information

The Contractor shall submit the following information on his proposals for the approval of the Engineer before work is commenced :-

- a) Source of supply of all aggregates
- b) Source of supply of cement
- c) Source of supply of water
- d) Grading curves for coarse and fine aggregate
- e) Certificate of soundness of cement
- f) Results of tests for organic and inorganic impurities of aggregates and water
- g) Results of tests of drying shrinkage of aggregates
- h) Source of supply of ready-mix concrete
- i) Source of supply of reinforcing steel
- j) Test certificates for reinforcing steel
- k) Types of water bars
- l) Types of joint fillers and sealants
- m) Positions of all construction joints
- n) Type of formwork

1.1.6 Use of Concrete

No concrete of any particular mix proportions shall be incorporated into the works until all the requirements of this specification have been met in respect of those mix proportions and have been approved by the Engineer.

1.1.7 Approvals

- a) Well before construction commences the Contractor shall supply to the Engineer for his approval details of his proposed layouts of concreting plant and on site workshop, details of formwork systems and construction devices, e.g. cranes, chutes, scaffolding, which he proposes to use for structural concrete. The information is to be sufficiently detailed to enable the Engineer to approve or otherwise.

- b) The Contractor should note that further approvals are required by the Specification before construction starts. The contractor is wholly responsible for obtaining these approvals and no claim for delays or additional payments will be entertained due to the contractor's failure to obtain such approvals in adequate time.

1.1.8 Responsibility and Defects

- a) The fact that the Contractor has used materials, etc. to the approval of the Engineer shall in no way relieve him of his responsibilities of producing a concrete of the required minimum strength, workability, shrinkage characteristics, etc. for the purpose to which it is put.
- b) Any defects due to materials and workmanship not in accordance with this specification shall be made good to the satisfaction of the Engineer by the Contractor at his own expense.

1.1.9 Source of Supply

The Contractor shall not alter the source of supply of any material from that initially approved by the Engineer without the written permission of the Engineer.

1.1.10 Supervision

The Contractor shall employ competent personnel upon the works whose first duty will be the supervision of all stages of making the concrete, site testing and preparation of cubes for works testing.

1.2 Concretor – Materials

All materials shall conform to "The Hong Kong Building (Construction) Regulations 1990" or in its absence the relevant current British Standard and shall be to the satisfaction of the Engineer. A copy of all relevant standards shall be provided by the Contractor and kept on site at all times.

1.2.1 Cement / PFA

Type of Cement

- a) Cement, unless otherwise specified, shall be ordinary or rapid-hardening Portland cement complying with BS 12.
- b) White or coloured Portland cement shall comply with the physical requirements of BS 12 and pigments shall be inorganic oxide pigments, either natural or synthetic in origin complying with the requirements of BS 1014. Pigments whether added just before mixing or by the cement manufacturer shall be incorporated in the proportions approved.

Certificate & Testing

- c) The contractor shall obtain a manufacturer's certificate of test in accordance with the appropriate standard for each consignment of cement delivered to the Site and shall immediately forward these certificates to the Engineer for his retention.
- d) Notwithstanding the manufacturer's certificate the Engineer may require that any cement delivered to the Site, or elsewhere for use in the Works, shall be sampled and tested. Any batch of cement so tested which fails to comply with this Specification will be rejected.

The Use of PFA

- e) The use of PFA shall conform to PNAP APP 33 and PNRC 13.

Delivery & Storage

- f) Cement of different types and manufacture shall not be mixed one with another. Cement shall be transported to site in covered vehicles adequately protected against water. All cement shall be delivered to the Site or elsewhere for use in the Works in the original sealed bags of the manufacturer or in approved bulk containers.
- g) Cement, unless delivered in bulk, shall be stored in a weather-proof shed, the floor of which shall be raised at least 300mm above the ground to allow free air circulation. Cement delivered in bulk shall be stored in a weatherproof silo. All cement shall at all times be protected from deterioration.
- h) Each consignment of cement shall be clearly identified, stored separately, and used in order of delivery.

Inspection

- i) Any cement which, upon inspection, is considered by the Engineer to have deteriorated in any way will be rejected.
- j) All cement rejected for any cause shall be removed from the site immediately.
- k) Bulk cement shall be used only if the Engineer's approval in writing has been obtained. Bulk cement shall conform in all respect to relevant Clauses above. Automatic weighing equipment shall be used for weighing the cement to be added to each batch.

1.2.2 Aggregates for Concrete

- a) Aggregates for concrete shall, unless otherwise specified be aggregates from natural sources complying with BS 882. Additionally, the flakiness index when determined by the sieve method described in BS 812 shall not exceed 35 for any size of concrete aggregate. Fine aggregate within or finer than zone 4 of BS 882 shall not be used. If there are any discrepancies between BS 882 and CS3, CS3 shall be adopted for the relevant parts.
- b) Aggregates shall not contain water soluble sulphur trioxide in excess of 0.1%.
- c) The chloride ion content of the fine and coarse aggregate shall not exceed 0.06% and 0.02% respectively by mass of dry aggregate. If either aggregate exceeds these limits the material may still be considered acceptable by the Engineer provided that the total chloride ion concentration derived from the aggregates is not greater than 0.35% by mass of the cement in the mix.
- d) Coarse aggregate shall mean aggregate of which not less than 95% by weight is retained on a 5mm mesh test sieve to BS 410. Maximum sizes of coarse aggregate are as specified in the Drawings.
- e) Fine aggregate shall mean aggregate of which not more than 10% by weight passes a no. 100 test sieve to BS 410 and 100% by weight passes a 10mm test sieve to BS 410.
- f) Between the limits specified for coarse and fine aggregates the grading shall be such as to produce a dense concrete of a consistency which will work into position without segregation and without the use of an excess water content.
- g) All aggregates shall comply with BS 882 and BS 812 in respect of impurities and additionally with the following:

Absorption	6% maximum for ordinary construction 2% maximum for waterproof construction
Salt	1% maximum (equivalent anhydrous calcium chloride percentage by weight of cement)
Shell	8% maximum
Carbonates	10% maximum

- h) The drying shrinkage of the aggregates measured in accordance with BS 812:Part 120 shall not exceed :
- 0.06% for general purpose concrete
- 0.04% for precast concrete and for concrete for special purposes, at waterproof concrete.
- i) Aggregates for waterproof concrete shall be non-porous not subject to internal movement, change in volume, shape or structure and suitable in every respect for use in waterproof concrete.

Approval

- j) Well before the commencement of any concreting work, the Contractor shall forward to the Engineer for approval details of his proposed source of supply of aggregates giving the aggregate group classification and typical physical properties as required by BS 882.

Certificate

- k) The Contractor shall provide the Engineer with a certificate for his retention at such regular intervals as the Engineer may prescribe from time to time showing that all aggregates have been re-tested since the date of the last such certificate and showing that they continue to comply with the requirements of this Specification.
- l) Samples of aggregates of different sizes or types shall be sent to an approved laboratory for testing prior to the commencement of any concreting works.
- m) Notwithstanding any certificate of compliance the Engineer may at any time require that any aggregate delivered to the Site, or elsewhere for use in the Works, be sampled and tested. Any aggregate so tested which fails to comply with this Specification will be rejected.

Delivery & Storage

- n) No deliveries in bulk shall be commenced until the samples have been approved by the Engineer as complying with this Specification.
- o) No deliveries in bulk shall be commenced until the samples have been approved by the Engineer as complying with this Specification.
- p) Coarse aggregate shall be delivered ready screened or screened on site into separate nominal single sizes within the limits given in BS 882.
- q) Aggregates of different sizes or type shall be stored in different hoppers or different stockpiles or approved well drained paved areas which shall be separated from each other such that mixing of the aggregates before batching is not readily possible. Stockpiles shall be protected against contamination from any source.
- r) Premixing of fine and coarse aggregates before delivery to site shall only be permitted with the written approval of the Engineer.
- s) Any aggregate, which in the opinion of the Engineer, has become contaminated or does not conform with the foregoing requirements may be rejected by the Engineer.

Test

- t) Tests shall be carried out on samples obtained from the delivered aggregates taken at intervals required by the Engineer in accordance with BS 812 Section 1 'Sampling of Aggregate'.
- u) The tests shall be those laid down in BS 812 Sections 2 to 6 inclusive. The tests shall be carried out in the presence of the Engineer/Inspector.
- v) Should a sample fail to comply with any of the tests the Inspector may, at his discretion, either reject the batch from which the sample was taken, order it to be washed and/or screened, or permit it to be used with variations in the proportions or the concrete mixes specified. Any batch of aggregate rejected by the Inspector shall be removed from site forthwith.
- w) Aggregates will be subjected to sieve analysis to BS 812, and the limits of acceptance are as shown in Table 1.1, 1.2 and 1.3. The Engineer may require further analysis to be made if there is any alteration in the type of aggregate.
- x) Provide a set of BS sieves and weighing equipment with accuracy to 0.5% of the mass of test sample and carry out sieve analysis to BS 812 on site as required by the Engineer.

TABLE 1.1
Grading of Coarse Aggregate

BS Seive (mm)	percentage by weight passing BS Sieve					
	Nominal size (mm) of graded aggregate			Nominal Size (mm) of single-sized aggregate		
	40 to 5	20 to 5	10 to 5	40	20	10
50.00	100	-	-	100	-	-
37.50	90-100	100	-	85-100	100	-
20.00	35-70	95-100	-	0-25	85-100	-
14.00	-	-	100	-	-	100
10.00	10-40	30-60	85-100	0-5	0-25	85-100
5.00	0-5	0-10	0-25	-	0-5	0-25
2.36	-	-	0-5	-	-	0-5

TABLE 1.2
Grading of Fine Aggregate

BS Sieve	percentage by weight passing BS Sieve		
	Grading Zone 1	Grading Zone 2	Grading Zone 3
10.00 mm	100	100	100
5.00 mm	90-100	90-100	90-100
2.36 mm	60-95	75-100	85-100
1.18 mm	30-70	55-90	75-100
600.00 um	15-34	35-59	60-79
300.00 um	5-20	8-30	12-40
150.00 um	0-10	0-10	0-10

TABLE 1.3
Grading of Crushed Stone Fine

BS Sieve	Percentage by weight passing BS Sieve
10.00 mm	100
5.00 mm	85-100
2.36 mm	60-87
1.18 mm	40-67
600.00 µm	20-50
300.00 µm	5-35
150.00 µm	0-20

1.2.3 Water for use with Cement

- a) Water for use in mixing with cement or for curing concrete shall be from Government main supply or any other approved source, clean, fresh and free from organic and other deleterious matter. Water storage tanks shall be covered to the satisfaction of the Engineer to prevent contamination.
- b) The Engineer may require that any water be sampled and tested by the method given in BS 3148. Water failing the criteria given in the appendix to BS 3148 will be rejected.
- c) Water for use in mixing with cement shall neither be hotter than 25°C nor colder than 5°C at the time of mixing.

1.2.4 Steel Bar Reinforcement

- a) Steel bar reinforcement shall consist of the following as described in the Drawings:
 - i. Mild steel bars complying with CS2:2012.
 - ii. Hot rolled high yield bars complying with CS2:2012.

Certificate

- b) The Contractor shall obtain certified evidence of the origin and a manufacturer's certificate of test in accordance with the appropriate standard for each steel batch relating to reinforcement delivered to Site and shall immediately forward these certificates to the Engineer for his retention.
- c) Sampling and testing of steel reinforcement shall be carried out in accordance with Practice Note for AP and RSE APP45. Testing shall be carried out by a HOKLAS accredited laboratory. Any reinforcement so sampled and tested which fails to comply with this Specification will be rejected.

Delivery & Storage

- d) All reinforcement shall be delivered to the Site either as straight bars or ready cut and bent to shape
- e) All reinforcement shall be stored at least 150mm clear of the ground or floor, in clean conditions and in an orderly manner to the satisfaction of the Engineer, such that the batch to which each piece belongs can be readily identified.

1.2.5 Steel Fabric Reinforcement

Steel fabric reinforcement shall be electrically cross welded steel mesh reinforcement complying with BS 4483.

1.2.6 Tying Wire

Tying wire for fixing reinforcement shall be either No. 16 gauge soft annealed iron wire, or No. 18 gauge stainless steel wires.

1.2.7 Spacers

- a) Spacer blocks for ensuring that the reinforcement is correctly positioned shall be as small as possible consistent with their purpose, of type and shape acceptable to the Engineer, and designed so that they will not overturn when the concrete is placed. They shall either be an acceptable proprietary brand of spacer, or they shall be made of concrete with 10mm maximum aggregate size and mix proportions to produce the same strength as the adjacent concrete.
- b) Spacer blocks of concrete shall not be used until at least 7 days old.

1.2.8 Admixtures

- a) No admixtures or cements containing additives shall be used in concrete unless specified or approved by the Engineer. Such approval will not be given unless in the Engineer's opinion specific benefit to the density or quality of the concrete will result. Additive approved shall be used strictly in accordance with the manufacturer's recommendations.
- b) The contractor shall obtain the written approval of the Engineer for use of additives if they differ from those specified by the Engineer, and their use is proposed by the contractor but not specified by the Engineer.
- c) Submit the following details of any admixtures or cement containing additives for approval before using :-
 - i. Brand name and manufacturer's literature;
 - ii. Proposed dosage and detrimental effects, if any, of underdosage and/or overdosage; and
 - iii. Method of adding to the concrete mix.
- d) Calcium chloride shall not be used under any circumstances.
- e) Where the use of admixtures is specified or approved they shall comply with the requirements of BS 5075.
- f) Store admixtures in watertight weatherproof containers clearly marked to show the contents.

1.2.9 Wall Ties

Wall ties shall be used between concrete and adjoining block or brick walling and shall be of an approved type.

1.2.10 Joint Fillers

Joint fillers shall be preformed and of a proprietary brand in accordance with the drawings or similar approved, and placed in accordance with the manufacturer's instructions.

1.2.11 Joint Sealants

- a) Joint sealants shall be as described in the Drawings and approved by the Engineer. Sealants shall be used strictly in accordance with the manufacturer's instructions.
- b) Poured joint sealing compound shall be a hot poured rubber bitumen compound complying with the requirements of BS 2499 unless otherwise specified.

1.2.12 Water Stops

Water stops shall be in accordance with the drawings or similar approved by Engineer.

1.2.13 Curing Compounds

- a) Aluminised curing compound shall, when applied at the rate of approximately 0.25 litres/m² by a mechanical sprayer with low pressure fine spray, become stable and impervious to evaporation of water from the concrete surface within 60 minutes after application.
- b) The curing compound shall contain sufficient flake aluminium in finely divided dispersion to produce a complete coverage of the sprayed surface with a metallic finish. The curing compound shall not react chemically with the concrete to be cured and shall not crack, peel or disintegrate within three weeks after application.
- c) Concrete curing compound for structures shall be a liquid resin or wax-resin base membrane curing compound of an approved proprietary brand. It shall have a minimum curing efficiency of 75% and shall contain a fugitive dye.
- d) Concrete curing compound for carriageways shall be a non-pigmented liquid resin or wax-resin base membrane curing compound of a proprietary brand. It shall achieve a 96% efficiency index and shall be available both with or without a fugitive dye.
- e) Supply test certificates, prepared by an approved testing laboratory, to show that the compound will provide the required curing efficiency.

1.3 Concretor – Workmanship

1.3.1 Reinforcement

Bending and Cutting

- a) The Contractor shall provide power plant necessary for cutting and bending of reinforcement on site accurately in accordance with BS 4466 to the shapes and dimensions shown on the Drawings or in the schedules where applicable. All reinforcement shall be bent at temperatures in the range of 5°C and 100°C. Bars shall be bent by the application of slow steady pressure.
- b) Cold worked or any high yield bars shall not be straightened or bent again once having been bent. When it is necessary to bend mild steel reinforcement already cast in the concrete the internal radius of such bends shall be not less than twice the diameter of the bar.
- c) All reinforcement shall be fixed in the position shown on the drawings by the adequate use of spacers, tying wire, chairs, stools, etc. and shall be so maintained during the concreting operation.
- d) Long vertical bars in walls and columns shall be adequately supported at a high level to ensure correct placement of the bars in lower lifts of concrete.
- e) Neither the size nor the length of a bars shall be less than the size or length shown in the drawings
- f) Where not specifically detailed in the Drawings, the laps of reinforcement shall be staggered so that laps do not occur in a band. Care shall be taken to comply with this requirement when using sliding formwork.
- g) Laps in reinforcement shall be where indicated on the Drawings or approved by the Engineer. Unless otherwise indicated the minimum lap length for steel bar reinforcement shall be 46 diameters and for mesh reinforcement 2 complete meshes.

Tolerances

- h) Unless specified otherwise by the Engineer, tolerances relative to dimensions for reinforcing steel shall be as follows :-
 - i. Bending Tolerances:-
 - 1) Cranks, stirrups and ties
 - For member depth less than 200mm +0, -5mm
 - For member depth 200mm or more +0, -10mm

- | | | |
|----|-------------|-----------|
| 2) | Other Steel | +0, -15mm |
|----|-------------|-----------|
- ii. Tolerances for Bar Spacing:-
- | | | |
|----|--|-------------------|
| 1) | Spacing of main bars in beams and columns | $\pm 10\text{mm}$ |
| 2) | Distance between layers of main steel | $\pm 5\text{mm}$ |
| 3) | Distance between bars along face of walls or slabs | $\pm 20\text{mm}$ |
- iii. Tolerances for Cover: +5mm and -0mm of the specified cover

Concrete Cover

- i) Cover for any reinforcement, unless specified otherwise on the Drawings, shall be as follows :-
- | | |
|---------------------------|---------------------|
| Water Retaining Structure | 40mm |
| Work below ground | 50mm |
| External Element | 40mm |
| Internal Element | 40mm |
| Slabs 20mm (ditto) | 20mm or dia. of bar |

Welding

- j) No welding of reinforcement shall be carried out without the approval of the Engineer.

Delivery, Storage and Testing

- k) Each consignment of reinforcement may be tested for compliance with the relevant British Standard Certificates or origin shall be produced.
- l) The Contractor shall bear all costs involved in making these tests. The Contractor shall have no claim for bars mutilated in obtaining test samples.
- m) The Contractor shall store reinforcement specified on site stock schedules in easily accessible and weather protected racks, capable of accommodating standard stock lengths.
- n) Approved mechanical splices shall be installed, where detailed on the drawings, in strict accordance with the manufacturer's instructions, unless specified otherwise.
- o) All splices shall be installed so that inspection is possible. Before concreting any member, where mechanical splices are used, the Contractor shall obtain the Engineer's approval of the splices installation. Unapproved installations shall be made good, at the Contractor's expense, to the satisfaction of the Engineer.
- p) All reinforcement shall at the time of concreting be free from mud, oil, mortar droppings, loose rust, paint, grease, mill scale or other deleterious matter. Reinforcement still 'blue' from the mill shall not be used.
- q) A steelfixer shall be in attendance at all times when concreting is in progress to correct any errors, omissions or movement in the reinforcement.
- r) In severe heat conditions reinforcement shall be shaded from direct sunlight and hosed down with clean water prior to concreting to keep the reinforcement below 25°C.

1.3.2 Formwork

- a) Formwork shall include all temporary or permanent requirement for forming the concrete, together with all temporary construction required for their support.
- b) All formwork shall be so constructed that there shall be no loss of material from the concrete. After the concrete has hardened and the formwork has been removed, the concrete shall be in the position and of the shape, dimensions, and surface finish described in this Specification or on the Drawings.
- c) In the case of beams, slabs or like members the formwork shall be so arranged that the sides or edges may be removed without disturbance to the soffit or propping system.

Formwork for External Wall Abutting Existing Building

- d) For the concreting for external wall abutting the existing building, the use of walls of adjoining buildings as permanent shuttering is not allowed. Except otherwise accepted by the Engineer, independent metal shuttering should be provided to prevent any load being imposed on the wall.

Precambering

- e) In general, and unless otherwise shown or specified in the Contract, all formwork for beams and slabs shall be set with an upward camber at midspan of 2mm per 1.0m of span for all spans in excess of 5.0m.

For cantilevers, camber 4mm per 1.0m at end of span.

Joints

- f) All joints in the formwork shall be either horizontal or vertical, end formwork being square across the mass of concrete. The design of the formwork shall be such as will permit of its being struck and moved without injury to the concrete.
- g) The Contractor shall use grout checks at all horizontal construction joints and shall adopt such other approved expedients, e.g. foam rubber sealing strips, polyester crack filler in joints, or taping of joints, as may be necessary to obtain satisfactory results at joints in formwork.

Metal Ties

- h) Where internal metal ties are permitted they or their removable parts shall be extracted without damage to the concrete and the remaining holes shall be carefully filled with mortar. No permanently embedded metal part shall have less cover than that indicated for adjacent steel reinforcement.

Holes

- i) When holes are to be provided in formwork for pipes, weep holes and the like they shall be neatly trimmed to fit the pipe and caulked with an approved material to form a grout-tight joint.

Sloping Forms

- j) When concrete is to be deposited to a steeper slope than 15° from the horizontal top forms shall be used to enable the concrete to be properly compacted.

Approval

- k) The Engineer may require details and/or calculations of any proposed formwork to be submitted for approval prior to work starting. Such approval if given shall not in any way relieve the Contractor of his responsibility for the safety or adequacy of such formwork for its purposes.

Coating

- l) The inside surface of forms, except for permanent formwork, or unless otherwise agreed by the Engineer, shall be coated with an approved material to prevent adhesion of the concrete. Such approved material shall be applied strictly in accordance with the manufacturer's instructions and shall not come into contact with reinforcement or other cast-in items.

Cleaning

- m) Immediately before concreting all forms shall be thoroughly cleaned out. In the case of deep sections an opening shall be left at the base to enable such cleansing to be adequately completed.

Inspection

- n) No concreting shall be commenced until the Engineer has had an opportunity to inspect the formwork and any enclosed reinforcement and given his approval to concreting.
- o) The erection, casing and striking of the formwork shall be done under the personal supervision of a competent foreman.

Striking of Formwork

- p) Formwork shall be struck at such a time and in such a manner as to cause no damage to the structure.
- q) The Contractor shall inform the engineer before he intends to remove any formwork. The time at which the formwork is removed shall be the Contractor's responsibility but the minimum periods for retaining formwork and falsework in position before striking shall be as follows and comply with Hong Kong Building (Construction) Regulation 61(2):-

Type of Formwork	<i>Minimum period before formwork may be removed</i>
Vertical formwork to foundation / footing	12 hours (w/o PFA) 30 hours (w/ PFA)
Vertical formwork to columns, walls and beams (unloaded)	12 hours (w/o PFA) 30 hours (w/ PFA)
Soffit formwork to slabs with props left in	4 days
Soffit formwork to beams with props left in	7 days
Props to slabs (unloaded)	10 days
Props to beams (unloaded)	14 days
Props to cantilevers (unloaded)	28 days

The periods given above are based on the removal of all props and formwork using ordinary Portland cement under average weather conditions. Adverse weather conditions or different cement may cause the above periods to be increased.

Where the props are to be left in place whilst the soffit formwork to slabs and beams is to be removed, such props shall either be designed to remain in position throughout the striking process without having to be temporarily removed or alternatively, if agreed by the Engineer, be so designed that a few can be removed at a time and replaced immediately.

No construction load is allowed to be imposed onto the unshored portion of structure under construction.

- r) Should the Contractor wish to reduce the striking times as specified in Clause 1.3.2(p) above, he must satisfy the Engineer and the Building Authority that the strength of the concrete at such time and the structural system is adequate to withstand the dead and imposed loads applied to it. The Contractor shall also at his own expenses submit the proposal with all necessary details and calculations endorsed by a RSE employed by the Contractor to the Engineer for the application for modification in accordance with Section 42 of the Building Ordinance from the provision of Building (Construction) Regulation 61(2).
- s) Where the structure is of multi-storey construction, props with head trees and braces shall be provided to distribute the imposed load below the floor being cast. This will normally be 2 storey heights below the floor being cast unless otherwise stated.
- t) When, in the opinion of the Engineer, the striking of formwork would be detrimental to or may prejudice the safety of the structure, the formwork shall be left in.

1.3.3 Formed Finishes for Concrete

- a) Where sawn formwork finish is specified or in all cases where no alternative finish is specified, the surface of the concrete shall be not worse than that obtained by the use of properly designed moulds of close jointed sawn boards. Small surface blemishes caused by entrapped air will be permitted but the surface should be free of voids, honeycombing or other defects.
- b) Where "fair faced" finish is specified the irregularities of the finish shall be no greater than those obtained from the use of wrought thickened square edged boards arranged in a uniform pattern. The concrete surface shall be smooth, free from fins, lippings, board marks or other irregularities, and even with sharp true rises. Only very minor blemishes or voids shall occur and there shall be no staining and no discoloration. The finish is intended to be left as struck and the only treatment generally permitted is light manual abrasion with a Carborundum stone to remove minor marks. Patching of any description will not normally be permitted.
- c) Where special finishes are required they will be described elsewhere in the Contract.
- d) Where a "fair faced" finish or other finish apart from "sawn formwork" finish is specified the Contractor shall provide a sample panel at least 2.4m x 1.2m in vertical surface area including a typical horizontal and vertical joint in the shuttering and the construction techniques that the Contractor proposes for the actual works. The sample when approved will form the standard for the entire Works. All unsuccessful samples shall be removed from the Site.
- e) Should any concrete in the Works, in the opinion of the Engineer, fail to match the standard sample in surface finish then the Engineer may for this sole reason order the removal of all such sub-standard concrete which shall be replaced with acceptable concrete all at the Contractor's expense.

1.3.4 Construction, Contraction and Expansion Joints

- a) The position and details of any construction joints not specified in the contract to be subject to Approval. Arrange such joints so as to minimise the possibility of the occurrence of shrinkage cracks.
- b) Vertical construction joints shall be properly made to form a vertical grout tight joint. Where reinforcement passes through the face of the joint the stopping off board shall be drilled so that the bars pass through, or the board shall be made in sections with half round indentations in the joint faces for each bar.
- c) Concrete when being deposited shall not be allowed to "tail off". Construction joints formed with expanded metal or similar will only be permitted for reinforced concrete work where approved.
- d) At all construction joints, both horizontal and vertical, the surface of the previously placed concrete shall be suitably treated to remove laitance and to expose the coarse aggregate thus forming an irregular surface key for adjacent concrete. This work shall be carried out to the satisfaction of the Engineer by the following or other approved methods: and compact as soon as practicable after deposition.
 - i. After the initial set has taken place but before final set the coarse aggregate shall be exposed by the use of water jet and stiff brushing.
 - ii. After final set has taken place the laitance shall be removed and coarse aggregate shall be exposed by bush hammering taking care not to loosen the aggregate in the face of the concrete matrix.

In either case the surface shall thoroughly washed and cleaned after roughening.

- e) At least 72 hours shall be left between completion of concreting one bay and the start of concreting any adjacent bay.
- f) Panels of walls and slabs not exceeding 40m long shall be cast alternately in such a sequence as to minimise and control the effect of shrinkage with construction joints. Shrinkage strips shall be introduced where necessary.
- g) Contraction joints shall be formed as detailed where shown on the Drawings.
- h) Expansion joints shall be formed as detailed at the positions shown on the Drawings.
- i) Waterstops to be provided at all construction joints in watertight structures.

1.3.5 Concrete Mixes - General

- a) Reference will be made to concrete mixes as Prescribed Mix concrete or Designed Mix concrete.
- b) The particular mix to be used will be as described in the Contract or on the Drawings.
- c) Mass concrete shall normally be Prescribed Mix concrete 15P unless otherwise specified.
- d) Blinding concrete shall normally be Prescribed Mix concrete 10P unless otherwise specified.
- e) The strength grade of concrete is denoted by the minimum characteristic strength in MPa with prefix "C"
- f) Grade of Prescribed Mix concrete is denoted by the characteristic strength in MPa and the nominal maximum size of aggregate in mm with suffix "P".
- g) Grade of Designed Mix concrete is denoted by the characteristic strength in MPa and the nominal maximum size of aggregate in mm with suffix "D" as specified in the Hong Kong Building (Construction) Regulations 1990.
- h) The characteristic strength of Designed Mix concrete with any type of cement, shall be the 28-days cube strength below which not more than 5% of the test results are expected to fall.
- i) The cementitious content is the mass of ordinary Portland cement (OPC) per cubic metre of the compacted concrete. If OPC is used in combination with PFA, the cementitious content is the combined mass of OPC and PFA.

The cementitious content of any mix for normal strength concrete ($f_{cu} \leq 60$ MPa) and high strength concrete ($f_{cu} > 60$ MPa) of the compacted concrete shall not exceed 550 and 450 kg/m³. For liquid retaining structures, the cementitious content shall not exceed 400 kg/m³.
- j) The workability of fresh concrete shall be at the discretion of the Contractor but subject to approval by the Engineer.
- k) The quality and strength of concrete shall be determined by test on samples taken on Site from the member during concreting and tested at an approved Laboratory as directed by the Engineer.
- l) The maximum size of aggregate for structural work shall generally be 20mm. For thin sections and where reinforcement is congested, 10mm aggregate may be approved or required. For large sections where the reinforcement is widely spaced and for mass concrete, 40mm aggregate may be approved or required.
- m) The free water / cement ratio is the ratio between the mass of the "free" water in the concrete mix and the cementitious content. The "free" water is that quantity of water available to combine with the cementitious materials. Any water required to be absorbed by aggregate is excluded. The free water/cement ratio will generally be chosen to suite strength and workability requirements but may be limited by the considerations of durability in accordance with Table 1.4.

TABLE 1.4

Maximum free water/cement ratio

Type of Structure Exposure	Moderate	Severe	Very Severe
Thin sections and sections with less than 25mm cover	0.50	0.45	0.40
Medium sections	0.55	0.50	0.45
Heavy sections, mass concrete and concrete placed under water	0.60	0.55	0.50

1.3.6 Concrete Mixes - Prescribed Mix Concrete

- a) Where Prescribed Mix concrete is required to be used the Contractor shall combine the cement, aggregate and water in the proportions given in table 1.5 to produce a concrete with high density combined with adequate workability for the purpose intended.
- b) The workability of the concrete shall be the minimum consistent with producing a dense, well compacted mass. Due regard shall be paid to the size and shape of the section together with any congestion of reinforcement.

TABLE 1.5
Prescribed Mix Concrete

Specified grade Strength	Material	Weight of aggregate per bag of cement kg	
		45kg bag of cement	50kg bag of cement
10 P	Fine aggregate	145	160
	20mm coarse aggregate	185	205
15 P	Fine aggregate	120	130
	20mm coarse aggregate	165	180
20 P	Fine aggregate	95	105
	20mm coarse aggregate	145	160

Weight of aggregate is measured when the aggregates are dry. Due allowance shall be made for the water content of moist aggregates.

1.3.7 Concrete Mixes - Quality A Concrete

Deleted.

1.3.8 Concrete Mixes - Designed Mixes

- a) All concrete arriving on site shall come from a concrete supplier registered under the Quality Scheme for the Production and Supply of Concrete (QSPSC).
- b) Mixes for each class of Designed Mix concrete specified or shown on the Drawings shall be designed by the Contractor to achieve the specified minimum cube strength combined with high density and adequate workability for the purpose. In order to allow for unavoidable variation the design strength shall exceed the specified characteristic strength by twice the expected standard deviation. In the absence of previous information a standard deviation of 7 N/mm^2 shall be assumed.
- c) Details of any proposed mix design shall be forwarded to the Engineer not less than 60 days before that class of concrete is required to be used on the works for his approval in principle. The details shall include at least the following information :-
 - i. Source, nature and grading of coarse and fine aggregates
 - ii. Source of cement and PFA
 - iii. Nominal maximum size of aggregate
 - iv. Cement and PFA content
 - v. Aggregate/cement ratio
 - vi. Free water/cement ratio
 - vii. Design density

- viii. Design slump or compacting factor
 - ix. Design strength
- In addition to the above, the following are to be submitted to the Engineer for concrete grade in excess of C45:
- x. Test reports on bleeding test
 - xi. Reports on heat of dehydration of the mix.
 - xii. Reports on creep and shrinkage deformation
 - xiii. Test reports on air content
 - xiv. Test reports on static modulus of elasticity at appropriate age
 - xv. The quality control procedures adopted by the batching plant with the CV of the quality control engineer
 - xvi. Local job reference
 - xvii. Detailed method statement of curing and time for formwork striking
- d) Classes of Designed Mix concrete will be referred to by the minimum 28 days Works cube strength.
 - e) The workability of the concrete shall be the minimum consistent which producing a dense, well compacted mass. Due regard shall be paid to the size and shape of the section together with any congestion of reinforcement.
 - f) The cementitious content of concrete shall be not less than the appropriate amount given in Table

TABLE 1.7

**Minimum cementitious content of Designed mixes (kg/m³)
(for Grade 20D and above)**

Exposure Condition	Max. Agg. Size (mm)	Plain Concrete	Reinforced Concrete
Moderate	40	220	260
	20	250	290
	10	300	340
Severe	40	270	320
	20	310	360
	10	360	410

- For aqueous liquid retaining structure, the minimum cementitious content to be 325 kg/m³, but subject to the limits set out in Table 1.7.
- g) Notwithstanding the requirements of free water cement ratio as stipulated under Table 1.4, for concrete grades 20D and above, the maximum free water cement ratio is not to exceed :-
 - i. 0.6 for unreinforced concrete
 - ii. 0.55 for reinforced concrete
 - iii. 0.5 for aqueous liquid retaining concrete
 - h) After the Engineer has approved a Designed Mix in principle, the Contractor shall prepare a trial mix on Site using the plant and materials intended for the Works. Each batch of Concrete shall be sampled and the following prepared, in accordance with CS1:2010.
 - i. Make one trial mix on each of three different days for each grade of concrete.

- ii. Prior to the sampling of test cubes, slump test (for design slump greater than 25mm) or compaction factor tests (for design slump less than 25mm) shall be conducted. The batch will be rejected should the slump test or compaction factor test fail to comply with the specification requirements.
 - iii. Take three samples from each mix and make two test cubes from each sample for testing at 28 days.
 - iv. The mix design shall not be approved if the average strength of the eighteen cubes tested does not exceed the specified strength by a margin of 12MPa, or the difference between the strengths of any two cubes from a single sample exceed 10% of the average strength of that pair.
- i) For concrete grade in excess of C45, in addition to trial mix as stated under Clause 1.3.8(g), the Contractor shall carry out a represented numbers of trial panels / trial columns to demonstrate the behaviour of the proposed concrete mix. The particulars as listed below are to be followed:
- i. The dimensions of the trial columns / panels shall be adequate to simulate the actual dimensions of the structural elements.
 - ii. Adequate numbers of thermocouplers shall be installed at appropriate locations to monitor the core temperature. The temperature monitoring shall be carried out continuously until the concrete temperature becomes steady.
 - iii. The workmanship for concreting works shall follow Clause 1.3.
 - iv. For each trial unit, 3 samples of concrete cubes and 2 test cubes from each sample shall be made for 7, 14 and 28 days for compression test.
 - v. Two concrete compression cores shall be taken from each trial unit for compression test.
 - vi. Should any of the following requirement not be achieved, the trial columns / panels are considered not acceptable. The Contractor shall review the mix or curing method and re-submit the details to the Engineer for approval and arrange for the re-test.
 - The concrete cube compression test result shall comply with Clause 1.3.8(g).
 - The concrete core compression test result shall comply with Clause 1.3.13.
 - The maximum temperature and temperature differential shall not exceed
 - 80oC and 25oC respectively.
 - There is no visual crack, segregation, honeycombing.

1.3.9 Ready Mixed Concrete

- a) Ready-mixed concrete shall be used only with the approval of the Engineer. When such approval is given the concrete shall be supplied in accordance with BS 5328 except where this conflicts with this Specification when in such case this specification shall prevail.
- b) Concrete shall be carried in purpose made agitators operating continuously, or in truck mixers.
- c) Ready-mixed concrete shall be compacted in its final position within 2 hours of the introduction of water to the mix, and within 30 minutes of discharge from the agitator or tuck mixer, provided that concrete which in the opinion of the Engineer is no longer sufficiently workable shall be rejected.
- d) The time of the introduction of water to the mix shall be recorded on the delivery note together with the weight of the constituents of each mix.
- e) When truck mixed concrete is used, water shall only be added under supervision either on Site or at the batching plant as agreed by the Engineer. Water shall not be added in transit.
- f) Truck-mixer units shall be maintained and operated in strict accordance with the manufacturer's recommendations.

- g) Delivery of ready-mixed concrete to site shall be carried out with prior approval of the supplier's production plant which shall be available for inspection by the Engineer during the supply period if required.
- h) A site record shall be maintained and made available for inspection at all times which contain the following information in relation to each delivery of concrete to Site:-
 - i. Registration number of truck, name of supplier and location of batching plant;
 - ii. Time of introduction of water to the mix;
 - iii. Time of arrival of lorry at the concreting location.
 - iv. Mix proportions, including admixtures, if any, and grade of concrete;
 - v. Position where concrete was placed;
 - vi. Whether test cubes were taken from the delivery, and details of cube markings; and
 - vii. Slump test results.
- i) The acceptance criteria for workability and strength shall be as specified in this specification.
- j) The use of ready mixed concrete shall not relieve the Contractor of any of his obligations and the appropriate clauses of this Specification shall apply equally to the ready mixed concrete.
- k) Concrete test cubes and slump tests shall be taken on the Site at the point and time of discharge in accordance with the Specification irrespective of any cubes that the supplier may take at his own works.
- l) When ready-mixed concrete is used, there shall be one supplier only and the supplier shall not be changed without the prior approval of the Engineer.

1.3.10 No-Fines Concrete

No-fines concrete shall be composed of ordinary Portland cement and 20mm - 10mm aggregate of which not more than 15% shall be retained on a 20mm BS sieve, and not more than 10% shall pass a 10mm BS sieve. The aggregate/cement ratio shall lie within the range 10 to 15 by weight and the cement content shall be such that each particle of aggregate is coated with cement paste but there is no continuous matrix. The water/cement ratio shall be 0.45 by weight unless agreed otherwise by the Engineer.

1.3.11 Mixing and Transporting Concrete

- a) All materials for concrete shall be measured by weight in approved weight batching equipment. Such equipment shall be checked at weekly intervals at the Contractor's expense and shall be accurate to within 2 per cent. Certificates of accuracy shall be submitted immediately to the Engineer.
- b) All concrete shall be mixed, unless otherwise exceptionally approved, in purpose built power driven mixers of a type and capacity suitable for the work and to BS 1305 or BS 4251. The mixers shall be equipped with accurate water measuring devices which shall be checked at weekly intervals at the Contractor's expense. Certificates of accuracy shall be submitted immediately to the Engineer. Mixers shall at all times be kept in a clean condition.
- c) Volume batching of materials and hand mixing of concrete will be allowed only at the discretion of the Engineer.
- d) Hand mixing of concrete shall be done on a clean close boarded platform. The materials shall be turned over not less than three times dry, water shall then be sprayed on through a hose and the materials again turned over not less than three times in a set condition and worked together until a mixture of uniform consistency is obtained.
- e) For hand-mixed or volume batched concrete the specified quantities of cement shall be increased by 10 per cent and not more than 0.5 cubic metre shall be mixed at one time. The Contractor shall not be entitled to any additional payment in respect of any extra cost of hand-mixing by reason of the additional cement or otherwise.

- f) All materials shall be thoroughly mixed dry before water is added and the mixing of each batch shall continue for a period of not less than two minutes after the water is added or such longer period as recommended by the manufacturer of the mixer. The mixture shall be of uniform colour and distribution of discharge and the entire contents of the mixer shall be discharged before recharging. The volume of mixed material shall not exceed the rated capacity of the mixer.
- g) Prior to the first mix each day being agitated in the mixer a rich cement/sand mix shall be used to coat the inside of the drum, the surplus material being emptied away and not used in the Works.
- h) A water/cement ratio shall be maintained to produce concrete of adequate workability. The workability shall be checked by using slump or compacting factor tests to CS1:2010. A guide to the degree of workability required for various types of work is shown in Table 1.8. For liquid retaining structures, minimum slump shall be 75mm.

TABLE 1.8

Workability

Degree of Workability	20mm max. size aggregate		40mm max. size aggregate		Uses of which Concrete is suitable
	Slump (mm)	Compacting factor	Slump (mm)	Compacting factor	
Low	12-25	.82-.88	25-50	.82-.88	Simply reinforced large sections with vibration
Medium	25-50	.88-.94	50-100	.88-.94	Moderately reinforced sections with vibration, such as ordinary beams and slabs
High	50-125	.94-.97	100-175	.94-.97	Sections with heavily congested reinforcement where vibration is difficult

- i) The moisture content of the coarse and fine aggregates shall be checked by the Contractor at frequent intervals and the amount of water added to the mix shall be adjusted to maintain the design workability.
- j) Concrete shall be discharged from the mixer directly into the works or onto a clean, level, watertight platform or into a clean watertight container. It shall be transported in a manner which ensures that it is of the correct quality and consistency at the point of deposition. All platforms and containers shall be clean of the old concrete before the fresh concrete is discharged onto them.
- k) No extra water shall be added to the concrete mix after it has left the batching plant.
- l) The Contractor shall take adequate precautions to protect concrete in transit from the effects of the weather.
- m) Pumping of concrete, which will require a special Designed Mix, will only be permitted with the approval of the engineer.
- n) Should the concreting be stopped due to mechanical malfunction, accident, or other similar cause then the Contractor shall inform the Engineer immediately so that necessary measures and precautions can be taken. The cost of any additional work caused by these stoppages shall be the responsibility of the contractor.
- o) No concreting shall be commenced until the formwork and reinforcement and all other preparations for the pour have been inspected by the Engineer.

1.3.12 Placing and Compacting Concrete

- a) Concrete shall be transported and placed by an approved method which ensures that there shall be no contamination, segregation or loss of the consistent materials.
- b) All areas where concrete is to be placed shall be clean and rendered free from standing water immediately before the placing of concrete, except for concrete placed under water.
- c) Concrete shall not be placed in any part of the structure until approval has been given; and if concreting is not started within 24 hours of that approval being given, approval shall be obtained again.
- d) Concreting shall proceed continuously between construction joints. Unless the Engineer agrees otherwise, fresh concrete shall not be placed against in-situ concrete which has been in position for more than 30 minutes unless the in-situ concrete has been kept damp with a layer of wet Hessian and protected from the weather.
- e) When in-situ concrete has been in place for 4 hours or in the opinion of the Engineer the concrete has stiffened to such an extent that it no longer responds to the action of an internal vibrator, no further fresh concrete shall be placed against it for a further 20 hours and construction joint shall be formed.
- f) Concrete shall be compacted in its final position within 30 minutes of discharge from the mixer unless it is carried in continuously operating purpose made agitators when the applicable time shall be within 2 hours of the introduction of water to the mix and within 30 minutes of discharge from the agitator. Concrete which in the opinion of the Engineer is no longer sufficiently workable shall be rejected.
- g) All concrete shall be vibrated unless otherwise specified. The vibration shall be carried out by experienced operatives using immersion type vibrators to the Engineer's satisfaction.
- h) Internal vibrators shall operate at not less than 10000 cycles per minute and external vibrators not less than 3000 cycles per minutes. Vibrating tables operating at a minimum of 5000 oscillations per minute may be used for precast elements subject to approval. Do not use external clamp-on type vibrators without prior approval.
- i) Placing of concrete shall be carried out in layers not exceeding 600mm deep and in sequence from one end of the form to the other.
- j) Concrete shall not be dropped from a height, thrown or otherwise treated so that segregation, undesirable finish, or defective structural quality results. Except otherwise agreed by the Engineer, the maximum free fall height shall not exceed 2.8m.
- k) Concrete in foundations and other underground work shall be protected from contamination by falling earth or rock during and after placing.
- l) Sufficient vibrators shall be provided to correspond with the rate of placing of concrete. The vibration shall be continuous throughout the placing of the concrete. Standby vibrators shall be on Site during all concrete placing.
- m) Vibration shall not be allowed to disturb any recently placed concrete that has begun to set. Any water accumulating on the surface of newly placed concrete shall be removed by approved means and no further concrete shall be placed thereon until such water is removed.
- n) Concrete shall be placed and compacted in such a way as to avoid disturbance to the formwork and reinforcement. Where sections of the Works are carried out in lifts, the reinforcement projecting above the lift being cast shall be supported to prevent movement of the bars during the casting and setting of concrete.
- o) Vibration shall not be used as a means of distributing concrete into position, and concrete shall not be vibrated between 4 and 24 hours after placing.
- p) Any concrete which in the opinion of the Engineer shows signs of initial setting before or during placing shall not be used and it shall be removed at the Contractor's expense.

- q) Suitable means shall be provided to ensure that the temperature of the concrete on placing does not exceed 30°C. Concrete shall not be placed around reinforcement or against surfaces which are at temperatures above 30°C. All surfaces shall be thoroughly dampened immediately prior to placing fresh concrete to prevent excessive absorption of water by the formwork.
- r) Underwater concrete shall be placed in accordance with the recommendations given in BS 8004.

1.3.13 Testing of Concrete

Concrete Cubes

- a) Except otherwise agreed by the Engineer, all structural concrete the following representative samples shall be taken, cured and tested in accordance with CS1:2010 on each day when concrete is being poured :
 - i. 150mm cubes - in accordance with the Hong Kong Building (Construction) Regulations 1990; and
 - ii. one slump test for each concrete lorry; or
 - iii. two compacting factor tests.
- b) The rate of sampling fresh concrete shall be at least that specified in Table 1.9 and at least one sample shall be taken from each grade of concrete produced on any one day.

TABLE 1.9
SAMPLING RATES

Type or part of building, building works or street	Quantity of concrete to be represented by each sample
Masts, cantilevers more than 3m in length, columns, shear walls and other critical elements	10m ³ or 10 batches whichever is the smaller volume
Solid rafts, pile caps, caisson caps and mass concrete retaining walls	100m ³ or 100 batches whichever is the smaller volume
All other types or parts	25m ³ or 25 batches whichever is the smaller volume

- c) All cubes shall be marked with the date of casting and a reference number. For each cube a record shall be kept of the position in which the batch of concrete from which it was sampled and placed.
- d) In addition to the water tank curing method specified in BS 1881, cubes may be cured on Site or in the laboratory in approved mist rooms. Where mist rooms are used, they shall comply with ASTM C511-75 except that the temperature shall be maintained at 27°C ± 3°C.

Testing and Test Record

- e) The testing and acceptance criteria shall be in accordance with the Hong Kong Building (Construction) Regulations 1990 and Code of Practice for Structural Use of Concrete 2013 unless more update requirements are specified by the Engineer.
- f) The concrete cubes tested at 28 days shall be taken to represent the concrete placed in the works. The standard of acceptance for cube strength tests shall be calculated from the maximum load sustained by the cube at failure.

Workability

- g) Where a batch is to be sampled to determine the workability of the concrete, samples shall be taken in accordance with either CS1:2010. In all cases the sample shall be remixed, divided into two specimens and each specimen tested for slump in accordance with CS1:2010.
- h) The standard of acceptance for the slump test during the production of concrete shall be within 25mm or 33% of the approved slump value, whichever is the greater.
- i) The standard of acceptance for the compacting factor test during the production of concrete shall be the design compacting factor plus or minus 0.03.

Testing

- j) Appropriately mark and deliver all concrete cubes and cores to an approved Laboratory, as directed by the Engineer.
- k) Indicate on the works test records the parts of the structure represented by the samples tested. Keep records on Site.
- l) The standard of acceptance of any preliminary tests will be similar to the standard for normal works cubes, slump or compacting factor, except the minimum cube strengths and standard deviation requirements as stated under Clause 1.38(g)(iv).
- o) The Engineer shall, during the progress of the work, have the power to order test, other than cube test, including slump tests in accordance with CS1:2010 as far as practicable due to local conditions.
- q) If the strength requirement is not satisfied the Engineer may order such further testing, including chemical analysis of samples of hardened concrete in the structure, as he considers necessary.
- r) Any concrete which fails to meet the above standards of acceptance shall be either further tested, or rejected, at the discretion of the Engineer. The costs of any such tests or of the removal of rejected concrete and its replacement and associated costs, shall be borne by the Contractor.
- s) If the strengths required are not attained or maintained throughout the Contract, the Engineer may instruct the Contractor to take any one of the following steps or a combination thereof as the Engineer considers appropriate :-
 - i. Adjust the mix proportion and carry out further trial mixes until a new mix proportion satisfying the requirements of the foregoing test is obtained and approved;
 - ii. Improve the standard of quality control;
 - iii. Carry out, at the Contractor's expense, other tests as directed by the Engineer;
 - iv. Replace any site concrete condemned as result of failing to comply with the standards of acceptance specified and meet all costs arising from such replacement. Submit for approval the method statement for replacing the condemned concrete before the replacement is carried out. In certain cases, the Engineer may require the method statement to be prepared and the remedial work to be supervised by a qualified structural Engineer.

Test Cores

- t) Where specified or when instructed by the Engineer, 100mm or 150mm diameter core samples from the finished concrete work shall be taken for 28 day test. Except otherwise specified in the coring scheme, cores may need to be taken at the locations decided by the Engineer.
- u) If concrete represented by test cubes in any concreting day fails to meet the standard of acceptance specified in accordance with the H.K. Building (Construction) Regulations 1990 or its latest edition, the Engineer may order the taking of 100mm or 150mm diameter core samples from the finished concrete work in same concreting day for test at the Contractor's expense. The locations of these cores to be taken shall be decided by the Engineer.

- v) All coring shall be completed within 7 working days of the date the instruction is given by the Engineer.
- w) All cores shall be drilled and tested in accordance with CS1:2010.
- x) Cored holes shall be made good to the approval of the Engineer with cement mortar or concrete of strength not inferior to that of the parent concrete.
- y) Results of the test cores for the same concreting day will be examined individually.
- z) In assessing the results of compressive tests on cores the concrete represented by the core shall be deemed not to comply with the specification if the strength of the core when adjusted for length / diameter ratio and converted to estimated cube strength in accordance with BS 1881 is as follows :-
 - i. The average result of the tested cores is less than 85% of the specified strength; or
 - ii. An individual result is less than 75% of the specified strength.
- za) The test core is considered not acceptable if the any of the following is observed:
 - The concrete core show evidence of segregation of individual materials.
 - The concrete core exhibit honeycombing which means interconnected voids arising from, for example, inadequate compaction or lack of mortar.
- zb) Cores shall not be tested for strength at ages less than 28 days and no adjustment shall be made to the measured strength in respect of the age of the core when tested.

1.3.14 Finishes for Concrete

Samples

- a) Provide samples of at least 1m of any exposed formed or worked finishes required, and obtain approval.

Fair-faced Finish

- b) Fair-faced finish shall be produced as follow :-
 - i. Obtain an even finish with a sheet material (e.g. plywood) free from fins, irregularities with true sharp arises.
 - ii. Arrange panels in pattern as specified or approved.
 - iii. The finish is to be left as struck. Do not make good until inspected by the Engineer. The only treatment generally permitted is light manual abrasion with a Carborundum stone. Patching will not nomrally be permitted.
 - iv. Blowholes not exceeding 10mm diameter will be permitted, but otherwise surface is to be free from voids, honey-combing or other large defects.
 - v. Variations in colour will only be permitted when the finished concrete is to receive and applied decoration.
 - vi. Make good small defects, and fill blowholes and formwork-tie holes with mortar specially mixed to match colour and texture.
 - vii. Abrupt irregularities shall not exceed 2mm.

Rough Board Finish

- c) Rough board finish shall be produced as follows :-
 - i. Formwork or formwork linings to consist of approved rough textured seasoned wood boards with moisture content of between 10% and 20%.
 - ii. Arrange boards of varying textures to give uniform overall effect.

- iii. Assemble boards to prevent penetration or loss of grout between them. Fill any gap greater than 5mm.
- iv. Soak formwork with water before erecting, and keep damp until concrete is placed.
- v. Position cover spacers and bolts to the satisfaction of the Engineer.
- vi. The number of uses of formwork lining will be limited to its ability to provide the required finish.
- vii. The finish shall be left as struck. Do not make good until inspected by the Engineer.
- viii. Fill minor voids and irregularities on surfaces with a matching mortar to approval, using a sponge rubber faced float.
- ix. Discoloration will not be permitted.

Worked Surface Finish

- d) Surface of concrete after casting shall be finished with one of the following surface finishes, level or to falls and currents as specified:-
 - i. Wood float to give an even textured surface.
 - ii. Steel trowel or power float to give a smooth untextured surface.
 - iii. Scratch with a stiff brush to give a slightly roughened even texture.
 - iv. Tamp with the edge of a board to give an even texture of parallel ribs.
 - v. Other finishes as specified and executed in an approved manner.
- e) Do not wet the surface to assist working and do not add cement to produce the specified finish without approval.

1.3.15 Curing and Protecting Concrete

- a) Immediately after compaction and for 5 days thereafter concrete shall be prevented from drying out too rapidly and protected against harmful effects of weather including rain, rapid temperature changes, and wind. The methods of protection used shall be subject to the approval of the Engineer. The method of curing used shall prevent loss of moisture from the concrete.

The above curing period shall be increased by 25% for concrete containing PFA.

- b) During the curing period horizontal surfaces shall be protected by one or more of the following or their approved means :
 - i. Covering with damp hessian canvas sacks or similar absorbent materials kept constantly damp and wholly covering the exposed concrete surface;
 - ii. Covering with an impermeable material raised approximately 50mm over the surface so as to prevent loss of moisture;
 - iii. An approved sprayed membrane curing compound.
- c) During the curing period other surfaces shall be protected by the following or other approved means :
 - i. Formwork in close contact with the concrete.
 - ii. Covering as in (b) above.
- d) All concrete faces or edges, particularly those which are exposed without rendering in the final structure, shall be adequately protected from damage and discoloration at all time.

- e) Concrete structures shall not be loaded until the concrete is at least 21 days old or 28 days in the case of cantilevers. With the prior approval of the Engineer the structure may be loaded after this time but in no case will loading be permitted.
- f) In no case the temperature differentials between the within the concrete mass and the external face shall exceed 25°C.

1.3.16 Tolerances

- a) All insitu concrete work shall be dimensionally accurate to within the following tolerances, unless closer tolerances are further specified or shown :

Except otherwise shown on drawings, the following are maximum allowable tolerances.

Item	Tolerance
1. Variations from the plumb,	
(a) In the lines and surfaces of columns, piers, walls and arises -	
In 3m	5mm
In any storey or 6m maximum	10mm
In 12m or more	20mm
(b) For exposed corner columns, control-joint grooves and other conspicuous lines -	
In any storey or 6m maximum	5mm
In 12m or more	10mm
2. Variation from the level, or from the grades indicated on the drawings *	At any point +/-10mm from true elevation but not more than 5mm variation between two points within 6m in a horizontal plane.
3. Variation of the linear building lines from established position in plan and related position of columns, walls and partitions:	
In any bay or 6m maximum	10mm
In 12m or more	25mm
No tolerance that would result in encroachment on adjoining properties is permissible	
4. Variation in the sizes and locations of sleeves, floor openings and wall openings	5mm
5. Cross sections of columns and beams, thicknesses of walls, slabs and other parts of the building:	
Up to 0.5m	+5/-3mm
Over 0.5m	+/-5mm
6. Footings -	
(a) Variation in dimensions in plan	+25mm# -10mm
(b) Misplacement or eccentricity: 2 percent of the footing width in the direction of misplacement but not more than	+25mm#
(c) Reduction in thickness	-5% of specified thickness

Item	Tolerance
7. Variation in steps -	
(a) In a flight of stairs :	
Rise	3mm
Tread	5mm
(b) In consecutive steps :	
Rise	2mm
Tread	3mm
* Variation in floor level are to be measured before removal of supporting shores.	
# Applies to concrete only, not to reinforcing bars or dowels.	

b) Closer tolerances will generally be specified for the following:

- (i) Precast concrete.
- (ii) Measuring flume contours.
- (iii) Filter bed and drying bed floors.
- (iv) Machinery bases.

1.3.17 Miscellaneous Items

- a) Holes, chases, indentations and the like shall be provided where indicated on the Drawings. All such shall be formed in the concrete and not cut after the concrete has hardened.
- b) Should the contractor or any sub-contractor require additional holes or the like these requirements shall be submitted to the Engineer for his approval prior to concreting.
- c) Pipes, conduits, fixing bolts and other such cast-in items shall be provided where indicated on the Drawings.
- d) Should the contractor or any sub-contractor require additional cast-in items these requirements shall be submitted to the Engineer for his approval prior to concreting.

1.3.18 Watertight Construction

- a) Concrete specified as "waterproof" is to contain an Approved water-proofing material mixed strictly in accordance with the manufacturer's instructions.
- b) Where watertight construction is specified, ensure that the work is free from leaks and damp patches. Basement including ground floor, roof and liquid retaining structure are in watertight construction unless otherwise stated.
- c) If a waterproof/vapour-proof membrane is to be applied, exercise special care to prepare the surface for the application in strict accordance with the manufacturer's instructions and to avoid damage to the membranes after application.
- d) The type of formwork-ties used is to leave no open holes through the concrete.
- e) Provide water stops for all construction joints unless approved otherwise. Well plan the positions of the construction joints according to daily production of concrete and keep to a minimum. If ready-

mixed concrete is used, supply to the site at regular intervals. The joint between the base slab and the walls to be minimum 250mm above the top of the base slab.

- f) Carry out remedial work required to eliminate any leaks and damp patches that occur, and obtain approval of method to be used.

1.3.19 Lintels

- a) Cast lintels in concrete Grade C30 either precast or cast in-situ, and construct as shown in Table 1.10.
- b) Provide 25mm minimum concrete cover between steel bar reinforcement and soffit.
- c) Allow bearing of 150mm (minimum) at each end.

TABLE 1.10
LINTELS

Clear span (m)	Depth of lintel (mm)	No. and diameter of mild steel bars per 105mm (or part) in width
0 - 1	150	one 12mm
1 - 2	225	one 16mm
2 - 3	300	one 20mm

1.3.20 Large Volume Pour

- a) All necessary precautions and measures shall be taken according to proven practices in concrete technology to minimise any form of cracking due to temperature and shrinkage effects in the casting of thick concrete sections (such as slabs, columns, walls, pile caps and footing /tie beams) whose least dimension exceeds 900mm.

Measures may include one or more of the following:

- i. Ground granulated blast furnace slag, PFA, or silica fume as a cement, substitute to reduce heat of hydration and to increase durability and water tightness of concrete.
 - ii. Other admixtures to increase workability, improve durability and reduce heat of hydration if necessary. The admixtures to be used in concrete shall not contain chlorides.
 - iii. Use of chilled water, flaked ice or liquid nitrogen as an additive to the concrete mix
 - iv. A coolant system to contain the thermal difference and minimise thermal cracking of concrete where necessary.
 - v. Controlled insulation of the concrete during curing.
 - vi. Delay in striking of formwork
- b) The rise in temperature within a period of 30 minutes shall not exceed 10°C, unless a means of controlling cracking has been employed which has been approved by the Engineer.
- c) The rate of subsequent cooling shall not exceed the rate of heating.
- d) The temperature differential between the warmer interior portion and the cooler surface portion of the thick element shall be limited to prevent early thermal cracking due to heat of hydration.
- e) The allowable differential shall take into consideration the boundary restraint of the structural element, and shall not exceed 20°C. The maximum temperature within the element shall not exceed 80°C for blended cement with 65% slag content, or 70°C where Portland cement only is used.

- f) A proposed method of placing, curing, temperature monitoring and any other measures shall be submitted to the Engineer for acceptance. This is a hold point. It shall be demonstrated that the proposal is adequate to prevent cracking arising from heat of hydration and drying shrinkage.
- g) The design concrete mix submitted for acceptance shall take the above requirements into account. all necessary tests are required to be performed to demonstrate to the acceptance of the Engineer, that the proposed mix design together with the proposals to limit temperature differentials and to prevent early thermal cracking are appropriate.
- h) Notwithstanding the above measures and precautions it is necessary, wherever practical, to seal, repair and rectify any cracks on any concrete elements with high strength cementitious mortars, epoxy resins or other accepted materials. If it appears that repair is impractical there may be a consideration for the possibility of redistributing the load on the affected element as an alternative to its demolition and reconstruction.

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